

Weather data:

This is weather data:

Result Grid Filter Rows: Export: Wrap Cell Content: Fetch rows:								
	Date/Time	Temp_C	Dew Point Temp_C	Rel Hum_%	Wind Speed_km/h	Visibility_km	Press_kPa	Weather
▶	1/1/2012 0:00	-1.8	-3.9	86	4	8	101.24	Fog
	1/1/2012 1:00	-1.8	-3.7	87	4	8	101.24	Fog
	1/1/2012 2:00	-1.8	-3.4	89	7	4	101.26	Freezing Drizzle,Fog
	1/1/2012 3:00	-1.5	-3.2	88	6	4	101.27	Freezing Drizzle,Fog
	1/1/2012 4:00	-1.5	-3.3	88	7	4.8	101.23	Fog
	1/1/2012 5:00	-1.4	-3.3	87	9	6.4	101.27	Fog
	1/1/2012 6:00	-1.5	-3.1	89	7	6.4	101.29	Fog
	1/1/2012 7:00	-1.4	-3.6	85	7	8	101.26	Fog
	1/1/2012 8:00	-1.4	-3.6	85	9	8	101.23	Fog
	1/1/2012 9:00	-1.3	-3.1	88	15	4	101.2	Fog
	1/1/2012 10:00	-1	-2.3	91	9	1.2	101.15	Fog
	1/1/2012 11:00	-0.5	-2.1	89	7	4	100.98	Fog
	1/1/2012 12:00	-0.2	-2	88	9	4.8	100.79	Fog
	1/1/2012 13:00	0.2	-1.7	87	13	4.8	100.58	Fog
	1/1/2012 14:00	0.8	-1.1	87	20	4.8	100.31	Fog
	1/1/2012 15:00	1.8	-0.4	85	22	6.4	100.07	Fog
	1/1/2012 16:00	2.6	-0.2	82	13	12.9	99.93	Mostly Cloudy
	1/1/2012 17:00	3	0	81	13	16.1	99.81	Cloudy
	1/1/2012 18:00	3.8	1	82	15	12.9	99.74	Rain
	1/1/2012 19:00	3.1	1.3	88	15	12.9	99.68	Rain
	1/1/2012 20:00	3.2	1.3	87	19	25	99.5	Cloudy
	1/1/2012 21:00	4	1.7	85	20	25	99.39	Cloudy
	1/1/2012 22:00	4.4	1.9	84	24	19.3	99.32	Rain Showers
	1/1/2012 23:00	5.3	2	79	30	25	99.31	Cloudy

weather 3 x

There are some problem to solve weather condition problems.

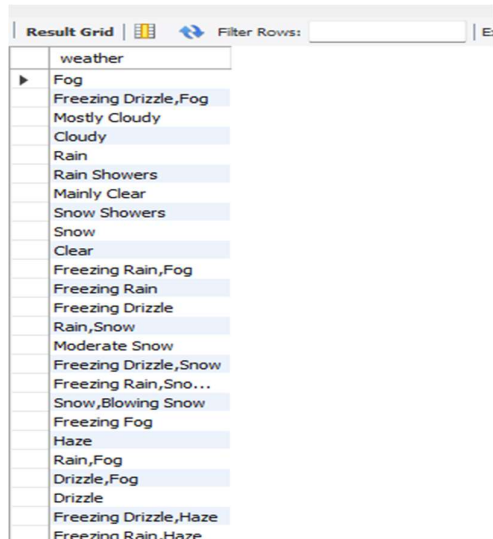
1.Find the maximum temperature recorded

Select max(Temp_C) from weather;

Result Grid Filter Rows:	
	max(Temp_C)
▶	33

2. Get distinct weather conditions

Select distinct weather from weather;

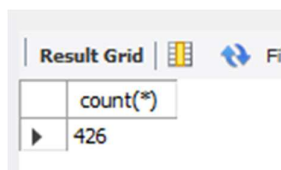


The screenshot shows a 'Result Grid' with a list of distinct weather conditions. The interface includes a 'Filter Rows' field and a 'Result Grid' tab. The list of weather conditions is as follows:

weather
Fog
Freezing Drizzle,Fog
Mostly Cloudy
Cloudy
Rain
Rain Showers
Mainly Clear
Snow Showers
Snow
Clear
Freezing Rain,Fog
Freezing Rain
Freezing Drizzle
Rain,Snow
Moderate Snow
Freezing Drizzle,Snow
Freezing Rain,Sno...
Snow,Blowing Snow
Freezing Fog
Haze
Rain,Fog
Drizzle,Fog
Drizzle
Freezing Drizzle,Haze
Freezing Rain,Haze

3. Write a query to find foggy hours.

select count(*) from weather where weather like '%fog%';

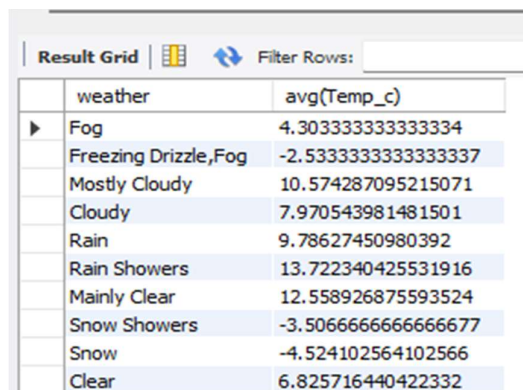


The screenshot shows a 'Result Grid' with the result of the query. The interface includes a 'Filter Rows' field and a 'Result Grid' tab. The result is as follows:

count(*)
426

4. Write a query to find average temperature by weather condition.

select weather,avg(Temp_c) from weather group by weather;



The screenshot shows a 'Result Grid' with the result of the query. The interface includes a 'Filter Rows' field and a 'Result Grid' tab. The result is as follows:

weather	avg(Temp_c)
Fog	4.303333333333334
Freezing Drizzle,Fog	-2.533333333333337
Mostly Cloudy	10.574287095215071
Cloudy	7.970543981481501
Rain	9.78627450980392
Rain Showers	13.722340425531916
Mainly Clear	12.558926875593524
Snow Showers	-3.5066666666666677
Snow	-4.524102564102566
Clear	6.825716440422332

5. Write a query to find number of hours with visibility below 1 km.

```
select count(*)
from weather
where Visibility_km<1;
```

Result Grid		Filter Rows
	count(*)	
▶	34	

6. Find the coldest hour(lowest temperature).

```
select `Date/Time`,Temp_C
from weather
order by Temp_C ASC
Limit 1;
```

Result Grid			Filter Rows:
	Date/Time	Temp_C	
▶	1/15/2012 8:00	-23.3	

7. Find the average pressure when temperature is below 0 c.

```
select avg(Press_Kpa)
from weather
where Temp_C<0;
```

Result Grid		Filter Rows:
	avg(Press_Kpa)	
▶	101.46307301293878	

8. count of weather conditions containing "Rain".

```
select count(*)from weather where weather like '%Rain%';
```

Result Grid		Filter F
	count(*)	
▶	689	

9. Find Top 5 most frequent weather conditions.

```

select weather,count(*) as frequency
from weather
group by weather
order by frequency desc
limit 5;

```

Result Grid   Filter Rows:		
	weather	frequency
▶	Mainly Clear	2106
	Mostly Cloudy	2069
	Cloudy	1728
	Clear	1326
	Snow	390

10. Find average visibility during foggy weather.

```

select avg(Visibility_km)
from weather
where weather like '%fog%';

```

Result Grid   Filter Rows:	
	avg(Visibility_km)
▶	6.024647887323955

11. Find the maximum dew point recorded when it occurred.

```

select `Date/Time`,`Dew Point Temp_C`
from weather
order by `Dew Point Temp_C` desc
Limit 1;

```

Result Grid   Filter Rows:		
	Date/Time	Dew Point Temp_C
▶	8/5/2012 15:00	24.4

12. Find the average wind speed for each weather condition.

```
select weather ,avg(`Wind Speed_km/h`)
```

```
from weather
```

```
group by weather;
```

Result Grid		
Filter Rows:		
Export:  Wra		
	weather	avg(`Wind Speed_km/h`)
▶	Fog	7.9467
	Freezing Drizzle,Fog	17.0000
	Mostly Cloudy	15.8139
	Cloudy	16.1273
	Rain	19.2549
	Rain Showers	17.1330
	Mainly Clear	14.1448
	Snow Showers	19.2333
	Snow	20.0385
	Clear	10.5573
	Freezing Rain,Fog	15.5000
	Freezing Rain	19.2143
	Freezing Drizzle	16.5714
	Rain,Snow	28.3889
	Moderate Snow	33.7500
	Freezing Drizzle,Snow	16.2727
	Freezing Rain,Sno...	32.0000
	Snow,Blowing Snow	34.8421
	Freezing Fog	4.7500
	Haze	10.4375
	Rain,Fog	14.7931
	Drizzle,Fog	11.8625
	Drizzle	16.0976
	Freezing Drizzle,Haze	10.3333